

Beyond Speed Rankings: Application-Layer Broadband Performance in Nine Southeast Asian Countries

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ABSTRACT

Southeast Asian broadband is usually judged by headline speed-index rankings, which say little about whether real applications work. Thailand ranked 12th worldwide for fixed broadband in June 2026, yet a 2023 national consumer survey found 81% of 2,924 respondents reporting connection problems. We analyze 990 million speed tests — 229.4M from Ookla Open Data and 760.6M from M-Lab NDT7, Q1 2023 to Q4 2025 — across nine countries, and score each province-quarter against the bandwidth requirements of voice, HD video, UHD video and cloud gaming. Our preliminary results show that basic connectivity is solved: voice and HD video succeed at 100% in every country, every quarter, on both fixed and mobile networks. That uniformity collapses at higher tiers. Fixed broadband stratifies sharply by geography, with a 16× spread between the strongest and weakest capital regions, and cloud-gaming success ranges from roughly 100% in Singapore to below 31% in Myanmar. Work is ongoing on per-province modelling and on peak-hour behaviour.

KEYWORDS

Internet measurement, broadband performance, Southeast Asia, Ookla, M-Lab NDT7, application requirements

1 MOTIVATION

National broadband quality in Southeast Asia is usually argued from two sources that disagree with each other: index rankings and user complaint. Thailand was ranked 12th worldwide for fixed broadband in Ookla’s Speedtest Global Index in June 2026 [5], while a 2023 survey by the Thailand Consumer Council found that 81% of 2,924 respondents had encountered connection problems, mostly slow or dropping connections. A national mean download speed cannot reconcile those two observations, because it describes the network rather than the service a user is trying to run over it.

We therefore ask a different question: *how do Southeast Asian networks perform against the demands of the applications people actually use?* Rather than ranking countries by speed, we score every province-quarter against published bandwidth requirements and report the share of measurements that clear each bar. The mismatch above is not unique to Thailand, so we run the analysis across nine countries.

2 DATA AND METHOD

We use two complementary open datasets covering Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia and Singapore, over twelve quarters (Q1 2023–Q4 2025):

- **Ookla Open Data** [4] — 229,367,221 tests, published as quarterly tiles of roughly 600 m×600 m, fixed and mobile.

Ookla servers sit inside access networks and use parallel streams, so these tests characterise last-mile capacity.

- **M-Lab NDT7** [3] — 760,607,387 single-stream tests to off-net servers, closer to what a single streaming or browsing flow observes.

Tiles are assigned to first-level administrative provinces by point-in-polygon against geoBoundaries ADM1 [6]; NDT7 points are joined to the same polygons. Province averages are test-weighted, and we keep only province-quarters with at least 100 tests and at least 5 tiles. Tests with non-positive throughput or duration are dropped. Because user-initiated speed tests are a biased sample [1], we report distributions and success shares rather than treating any single figure as a population mean.

We evaluate four application classes against the thresholds of Lübben and Misfeld [2]: voice (64 kbps), HD video streaming (5 Mbps), UHD video streaming (25 Mbps) and cloud gaming (44 Mbps). **Success rate** is the national share of measurements meeting the *download-speed* threshold in a given quarter. We deliberately do not evaluate the 25 ms latency component of the cloud-gaming requirement: M-Lab servers are off-net for eight of our nine countries — Thai tests predominantly terminate in Singapore, and only the Philippines is served domestically — so measured round-trip time reflects international transit rather than domestic access.

3 PRELIMINARY RESULTS

3.1 Basic connectivity is solved

Voice and HD video succeed at 100% in every country, in every quarter, on both fixed and mobile networks. This holds for the lowest-tier markets in the set as firmly as for Singapore. Whatever the regional digital divide consists of, it is not a failure to carry a phone call or a 5 Mbps video stream.

3.2 Fixed broadband stratifies by geography

Most capital regions outperform their own country (Figure 1): Bangkok reaches 298.8 Mbps against a national 276.9, Kuala Lumpur 212.3 against 194.8, and Manila NCR 159.6 against 132.0. Singapore is the instructive exception — its Central Region (361.0) sits *below* the national mean (394.8), because a city-state has no rural hinterland to average against. The spread between the strongest and weakest capital region is over 16× (Singapore against Yangon at 22.0 Mbps), a gap that a table of national averages conceals.

The trajectory is also diverging rather than converging. Singapore nearly doubled its median provincial fixed speed over the period, from roughly 280 to 550 Mbps, and Thailand grew from roughly 200 to 300. Vietnam accelerated to join Malaysia above 200 Mbps. Laos, Cambodia, Indonesia and Myanmar remained below 60 Mbps for all three years.

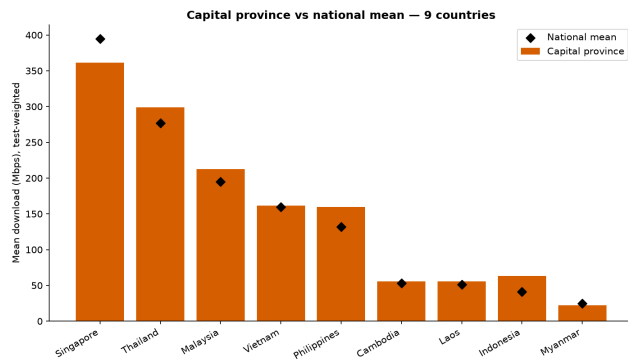


Figure 1: Capital province versus national mean download speed, fixed broadband, test-weighted, nine countries.

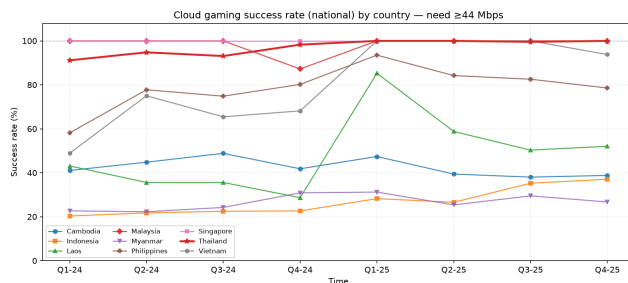


Figure 2: Cloud-gaming success rate (≥ 44 Mbps download) by country and quarter.

Fixed and mobile do not stratify alike. Fixed broadband exceeds mobile by $3.2\times$ in Thailand and $2.1\times$ in Singapore, but the two are near parity in Malaysia and Indonesia ($1.1\times$), and Myanmar is the only country in the set where mobile is *faster* than fixed ($0.8\times$) — a mobile-first market in which fixed investment has not followed.

3.3 Application tiers diverge

Raising the bar from HD to UHD video separates the region into two groups. Malaysia, Singapore, Thailand, the Philippines and Vietnam clear the 25 Mbps threshold in 100% of measurements every quarter, while Cambodia ranges from 67% to 86%, Indonesia improves from 36% to 65%, and Myanmar stays between 40% and 55%. Laos is volatile across the full 50–100% range, which we attribute to its small measurement volume rather than to genuine quarter-to-quarter swings in capacity.

Cloud gaming separates them further (Figure 2). Singapore sustains approximately 100%, and Thailand and Malaysia hold 87–100%. The Philippines ranges from 58% to 93%, and Vietnam climbs from under 50% to roughly 94% over the period. At the other end, Cambodia reaches only 38–49%, Laos approximately 52%, Indonesia stays below 38% and Myanmar below 31%. The same networks that serve every user adequately for voice fail the majority of their users for a latency- and bandwidth-sensitive interactive application.

4 ONGOING AND FUTURE WORK

Three directions are in progress. First, we are fitting province-level models of speed against economic and density covariates to separate infrastructure effects from wealth effects. Second, we are extending the NDT7 analysis to peak-hour behaviour, where busy-versus-quiet *ratios* remain interpretable even though absolute latency is not, given off-net server placement. Third, we are quantifying the measurement bias introduced by thin-sample provinces, which currently inflate the apparent performance of several emerging markets.

The broader argument we wish to discuss at the poster session is that a speed index answers the wrong question. Basic access across Southeast Asia is effectively universal; what separates these markets is whether their networks survive contact with a demanding application, and that distinction points at different infrastructure investment than a ranking does.

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